SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	Cse AIML	Date:	

Code of Conduct

S Santhosh Reddy (192425017)_(Name / Reg No)certify that this submission is my original work and that I have adheredtotheguidelinesspecifiedforthisassessment.Iunderstandthatanyviolationofacademicintegrityruleswillresult in disciplinary action.

SignatureoftheStuden:S Santhosh Reddy

Instructions

1. Readallquestionscarefullybeforeattemptingtheanswers.
2. Answerallfivequestions.Eachquestioncarriesequalmarks.
3. Showallintermediatesteps,calculations,probabilitycomputations,taggingdecisions,andjustifications wherever applicable.
4. Duration:30minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as "analyzing", "analysis", and "analytical". The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

Answer

Morphological Processing Approach

Morphological processing analyzes words by separating them into:

- Root (Stem): The main meaning of the word.
- Affixes: Prefixes or suffixes added to the root. Steps

1. Read the input word.
2. Identify the root and suffix/prefix.
3. Check whether the suffix is:
   - o Inflectional (changes grammar only)
   - o Derivational (creates a new word)
4. Convert all related words to a common root (analyze) for indexing.
5. Store the normalized root in the search index.

Morphological Analysis

Input Word	Root	Affix	Type	Normalized Form
Analyzing	Analyze	-ing	Inflectional	Analyze
Analysis	Analyze	-sis	Derivational	Analyze
Analytical	Analyze	-ical	Derivational	Analyze

Morphological Structure

- Analyzing = Analyze + ing
- Analysis = Analyze + sis
- Analytical = Analyze + ical

Final Result

All three words are normalized to the common root:

Analyze

This allows the search engine to retrieve the same relevant documents even when users search using different word forms.

Conclusion

Morphological processing improves search accuracy by:

- Extracting the root word.
- Identifying inflectional and derivational suffixes.
- Normalizing all related words to a single root (Analyze) for efficient indexing and retrieval.

2. An AI-based sentiment analysis platform receives product reviews containing words like "disagree", "agreement", and "agreeable". The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

Answer (Simple and Clear)

Morphological Parsing Strategy

The system processes each word by:

1. Identifying the base (root) word.
2. Detecting any prefixes and suffixes.
3. Classifying the word formation as derivational or inflectional.
4. Understanding how the added prefix or suffix changes the meaning.
5. Normalizing related words to the base form (agree) for better sentiment analysis.

Morphological Analysis

Input Word	Prefix	Root	Suffix	Transformation Type	Meaning
Disagree	dis-	agree	—	Derivational (Prefix)	Opposite of agree (negative)
Agreement	—	agree	-ment	Derivational (Suffix)	State or act of agreeing
Agreeable	—	agree	-able	Derivational (Suffix)	Pleasant or willing to agree

Morphological Structure

- Disagree = dis + agree
- Agreement = agree + ment
- Agreeable = agree + able

Root Form Agree Conclusion
The system identifies "agree" as the common root, recognizes prefixes and suffixes, and understands how derivational changes affect meaning. This helps the sentiment analysis system correctly interpret positive and negative sentiments across different word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as "govern", "government", and "governance". The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

Answer (Simple and Clear)

Morphology-Based Normalization Mechanism The system performs the following steps:

1. Read the input word.
2. Identify the root (base word).
3. Separate any suffixes.
4. Determine whether the suffix is derivational or inflectional.
5. Normalize all related words to the common base form (govern) for topic modeling and clustering. Morphological Analysis

Input Word	Root	Suffix	Transformation Type	Normalized Form
Govern	govern	—	Base Word	Govern
Government	govern	-ment	Derivational	Govern
Governance	govern	-ance	Derivational	Govern

Morphological Structure

- Govern = govern
- Government = govern + ment
- Governance = govern + ance Root Form

Govern Conclusion

The system extracts the common root "govern", identifies the derivational suffixes -ment and -ance, and normalizes all words to "govern". This improves topic modeling, document clustering, and search accuracy by treating related word forms as the same concept.

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

Answer

Morphological Parsing Strategy

The system performs the following steps:

1. Read the input word.
2. Identify the root (base word).
3. Detect any prefixes and suffixes.
4. Classify them as derivational or inflectional.
5. Analyze how they change the meaning and word class.
6. Normalize all related words to the common base form "activate" for classification and indexing.

Morphological Analysis

Input Word	Prefix	Root	Suffix	Transformation Type	Normalized Form
Activate	—	activate	—	Base Word	Activate
Activation	—	activate	-ion	Derivational	Activate
Reactivation	re-	activate	-ion	Derivational (Prefix + Suffix)	Activate

Morphological Structure

- Activate = activate
- Activation = activate + ion
- Reactivation = re + activate + ion Root Form

Activate Conclusion

The system extracts the common root "activate", identifies the prefix re- and the suffix -ion, and normalizes all words to "activate". This improves document classification, semantic grouping, and search indexing by treating related word forms as the same concept.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning. Apply the process to extract root forms and classify each transformation. Determine the morphological breakdown and normalized output for each word.

Answer (Simple and Clear)
Morphology-Based Normalization Approach The system performs the following steps:

5. Read the input word.
6. Identify the root (base word).
7. Detect the inflectional suffix.
8. Classify the grammatical change (tense or aspect).
9. Remove the suffix and normalize all words to the common base form "create".

Morphological Analysis

Input Word	Root	Suffix	Transformation Type	Normalized Form
Create	create	—	Base Word	Create
Creates	create	-s	Inflectional (Present Tense, 3rd Person Singular)	Create
Creating	create	-ing	Inflectional (Present Participle/Continuous Aspect)	Create

Morphological Structure

- Create = create
- Creates = create + s
- Creating = create + ing

Root Form Create

Conclusion
The system extracts the common root "create", identifies the inflectional suffixes -s and -ing, and normalizes all words to "create". This ensures consistent search results, better indexing, and accurate retrieval by treating grammatical variations as the same word.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date:	Date:	Date: